

method would result in a complete circuit, and it also allows some circuit elements to be added easily—an edge over the other solvers. This type of solver allows analysis both in time and frequency domain.

Methods using the differential Maxwell equations

Finite-difference time domain (FDTD).

It is a time domain technique and thus a single simulation can cover a wide frequency range. This method allows easy computations for each cell made of different materials. After the process, the time domain results are converted to frequency domain by the use of DFT techniques.

Consider a large structure. This method will not produce dense material matrices (as done by others); it will include only the nearest neighbour interaction. This method can cover frequencies from near low to microwave, and up to visible light frequencies. The discretisation grid used is rectangular (usually) and separate grids are used for E and H calculations.

A disadvantage of the method is that, although one simulation will work for a wide frequency range, this is not the case with the ports. A single simulation has to run for each port placed on the geometry. So, if there are N ports in a structure, N simulations would be required.

Some applications of the method are:

1. Time domain reflectometry (TDR)
2. Antenna placement on cars, trains, etc
3. Simulation of phone with SAM head to calculate SAR value
4. When the module is electrically large and ports are less

The other side of the story

Although CEM results are quite accurate for the complex problems but they cannot be relied on completely. A particular method is used to solve particular type of problems.

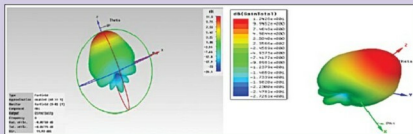


Fig. 8: Validating gain pattern of horn antenna. Mark the differences between CST (left) and HFSS (right)

These methods also have errors associated with them. Thus there is a strong requirement for validation of the results. Validation is done to answer the question "How much does the result obtained by numerical method resemble the reality?"

Validation techniques

In validation process, basically a comparison is done between the patterns, values, etc. There are four approaches to validating the results of a numerical method:

Validating using other numerical solution. Different numerical methods with proper modelling of problem can be used to validate, as shown in Fig. 8.

Validating using analytical solution. If the analytical theory behind the problem is known then that can be used to validate the results of the simulation.

Validating using experimental results. By performing the experiment of the problem under the same condition.

Validating using intermediate results. Intermediate results of the simulation and the experimental results are cross-checked to save the length of validation.

Finite element method (FEM). It is defined as an approximation technique to discretise geometric objects. Each part of discretised structure is called a finite element. It is a frequency-domain flexible method for geometrical and material handling of closed structures. It generates a sparse matrix of unknown field quantities. This method requires meshing of entire space and needs additional ABC boundary over the structure to be analysed. The unknown field quantities are approximated over each mesh as sum of functions, as shown in Fig. 8.

Its disadvantage is that, for simulation of structures of complex and elec-

trically large geometries, the mesh may become very complex. And that would result in large matrices which would require huge memory resources.

Its advantages are:

1. It is most suitable for high 'Q' circuits
2. Useful for analysis of filters, cavities, resonators, etc
3. Provides good results for geometries with large number of ports, such as IC packages and multi-chip modules.

Apart from these, several other methods like FIT (finite integration method) over FEM, TLM over FDTD, MLFMM over MOM have been de-

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